

HARINANDAN REDDY SOMA

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Professional Summary

Results-driven Senior Data Scientist with 5 years of experience specializing in Artificial Intelligence, Machine Learning, and MLOps engineering. Proven expertise in designing, implementing, and optimizing end-to-end machine learning solutions across various industries, including finance, healthcare, digital marketing, manufacturing, e-commerce, and technology. Adept at leveraging advanced statistical models, deep learning techniques, and big data technologies to drive business insights and innovative solutions. Strong background in developing scalable, production-ready ML pipelines using cloud platforms and MLOps tools, ensuring seamless deployment and monitoring of models. Excellent problem-solving skills, combined with a passion for continuous learning and staying ahead of industry trends. Demonstrated ability to lead cross-functional teams, mentor junior data scientists, and collaborate with stakeholders to translate complex data-driven insights into actionable strategies that enhance business performance and drive growth.

Skills

- Programming Languages: Python, SQL, JavaScript
- Machine Learning & AI: Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, OpenCV, NLTK, Spacy, Supervised, Unsupervised, Recommendation systems and Reinforcement Learning, Classification, Regression, Clustering, Dimensionality Reduction, Object detection, Image Classification, Optical Character Recognition (OCR), Text classification, Prompt Engineering
- Deep Learning: Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Generative Adversarial Networks (GANs), Transformer Models, NLP (Natural Language Processing), Computer Vision - YOLO, Detectron2, Tesseract, Large Multimodal Model (LMM) – Llava, Automatic Speech Recognition – Whisper, Kaldi
- Generative AI: Large Language models (LLMs): Llama, OpenAI GPT - 4, Claude, PEFT, LoRa, QLoRA, zero-shot, one-shot, few-shot learning, Fine-tuning, Retrieval Augmented Generation (RAG) Langchain, Llama Index, Vector Databases and Chatbot development
- Statistical Analysis: Pandas, NumPy, SciPy, A/B testing, hypothesis testing, t-test, chi-square tests, ANOVA, Time Series, Correlation and Covariance Analysis
- Web Development: Flask, FastAPI, Django, ray, PySpark, Streamlit, React JS, RESTful APIs, CRUD Operations, User Interface (UI/UX) design, Asynchronous, Multi-threading, Multi-processing, Concurrent, Parallel Programming and distributed computing
- MLOps Tools: MLflow, Kubeflow, Airflow, Containerization - Docker, Load balancing, Container Orchestration - Kubernetes, Pods, Jenkins, Git, CI/CD pipelines

- Cloud Platforms: AWS (SageMaker, Lambda, ECS, EKS, EC2 Redshift, S3), Azure (ML Studio, Databricks, Azure DevOps), GCP (AI Platform, BigQuery, Dataflow)
- Databases: MySQL, PostgreSQL, MongoDB, Cassandra, Redis, Elasticsearch
- Visualization Tools: Tableau, Power BI, Matplotlib, Seaborn, Plotly, D3.js
- Operating Systems: Linux, Unix, Windows
- Fluent Business English communication and Stakeholder management

Work Experience

Senior Data Scientist, 12/2023 to Current, **XFactr.AI, Client - Schnieder Electric** - Chennai

Technologies: Python, TensorFlow, PyTorch, MLflow, Kubernetes, Apache Spark, AWS (SageMaker, Redshift), Git, Jenkins, Docker, SQL, Tableau

AI Developer, 09/2023 to 12/2023, **VMG Digital** - Chennai

Technologies: Python, Keras, OpenCV, TensorFlow, Airflow, Docker, Azure (ML Studio, Databricks), SQL, Power BI

Machine Learning Engineer, 05/2022 to 03/2023, **3i Infotech** - Chennai

Technologies: Python, Scikit-learn, XGBoost, Apache Hive, Kafka, GCP (AI Platform, BigQuery), GitLab, CI/CD, SQL

Senior Client Partner – Data Science, 04/2021 to 10/2021, **Access Healthcare** - Chennai

Technologies: Python, Scikit-learn, Hadoop, Hive, SQL, Tableau, Power BI, Git

Executive – Data Science, 03/2020 to 04/2021, **Omega Healthcare** - Chennai

Technologies: Python, TensorFlow, Keras, SQL, Tableau, Git, Jenkins

Associate – Data Science, 11/2017 to 12/2018, **Vertex Solutions** - Chennai

Technologies: Python, Scikit-learn, SQL, Tableau, Git

Projects

Personal Protection Equipment (PPE) Monitoring System

Developed a comprehensive AI-based PPE Monitoring System to ensure workplace safety by detecting and verifying workers' use of personal protection equipment (PPE) in real-time. The system utilizes advanced computer vision techniques and deep learning models to analyze video feeds from cameras installed in industrial environments, automatically identifying whether workers are wearing required PPE such as helmet, gloves, mask, boots, goggles, and safety vest. The PPE Monitoring System successfully reduced workplace safety incidents by 90% within the first three months of deployment. It provided real-time compliance tracking, significantly improving the overall safety culture within the organization

Automated Quality Assurance System for Contact Centers

Designed and implemented an advanced Quality Assurance (QA) Automation System for contact centers, utilizing Natural Language Processing (NLP), Large Language Models (LLMs), and Automatic Speech Recognition (ASR) technologies. The system automates the evaluation of customer-agent interactions by transcribing, analyzing, and scoring conversations, significantly reducing the manual workload of quality agents while ensuring consistent, accurate, and real-time assessments. The Automated Quality Assurance System led to an 80% reduction in manual QA efforts, increased the consistency and accuracy of agent evaluations, and provided real-time feedback, improving overall agent performance. The system also enhanced customer satisfaction by enabling prompt identification and resolution of issues, contributing to a 30% improvement in customer satisfaction scores.

Automation of conversion of Static Digital Catalogs into Dynamic Digital Catalogs

Led the development of a sophisticated automation system that converts static digital catalogs (e.g., PDFs, scanned images) into dynamic, interactive digital catalogs using Optical Character Recognition (OCR) and Computer Vision. This project aimed to streamline the catalog updating process, allowing businesses to maintain up-to-date, user-friendly digital catalogs with real-time data and interactive features. The project successfully automated the conversion of static digital catalogs into dynamic, interactive catalogs, reducing manual processing time by 60%. The system's ability to automatically update and personalize catalog content based on real-time data resulted in a 30% increase in user engagement and a 25% improvement in catalog management efficiency. The dynamic catalogs also enhanced the customer experience by providing up-to-date, easily navigable content.

Education

M.Sc. in AI: Pursuing, **IU** – Germany

B.Sc. Computer Science: 2019, **Yogi Vemana University** – Kadapa

12th: 2016, **Government Junior College** – Pulivendla

10th: 2014, **Venkatappa Memorial School** – Pulivendla

Certifications

- Python
- Machine Learning
- Git and GitHub
- Agile Scrum Master
- SQL
- Mathematics
- Statistics
- DevOps